Mixture of Experts in Zero Shot Transfer of Legal Judgement Prediction as Article-aware Entailment for the European Court of Human Rights

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ABSTRACT

In the Legal Judgement Prediction (LJP) entailment task, the goal is to classify the outcome of a legal case based on a combination of case facts and convention articles. By evaluating a model's ability to classify legal cases based on articles it has never seen before, the Legal Judgement Prediction (LJP) entailment task provides a way to test the model's zero-shot ability. This task was introduced in a previous paper that also proposed a transformer-based entailment model to solve different zero-shot scenarios on the European Court of Human Rights (ECtHR) dataset. They observed that domain adaptation techniques such as the domain discriminator can improve their models' performance, especially for unknown articles. Additionally, they found that a model trained on articles related to a particular unseen article can perform better on that article compared to a model trained on unrelated articles.

In this paper, we propose a new approach to address this challenge with Mixture of Expert (MoE) models. The goal of this paper is to show that our MoE models can increase the performance of the previously introduced entailment model with domain discriminator. Our models combine the predictions of experts for each known article and a general model. The general model is equivalent to the entailment model with domain discriminator and each expert equivalent to the entailment model. The goal of our proposed MoE-Attention model is to learn to weigh experts in a beneficial manner depending on the input article of the entailment task. We are able to outperform the entailment model, but not the entailment model using a domain discriminator. However, we believe this could be improved with further experimentation and hyperparameter tuning. Our findings suggest that we can learn to weigh experts in a beneficial manner even for unknown articles. At this point in time, our results suggest that MoE is not able to improve the zero-shot ability of the underlying entailment models.

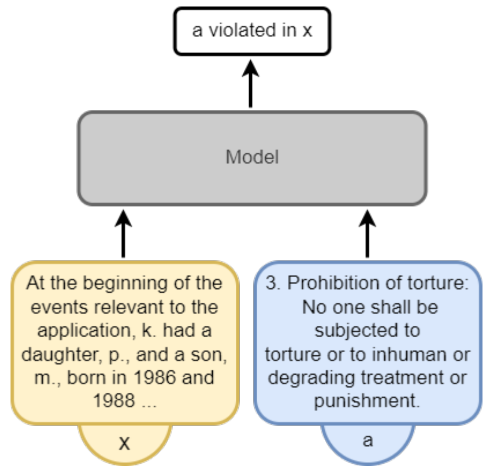

Figure 1: Entailment task example

1 INTRODUCTION

With the rise of automated legal systems, the ability to predict the outcome of legal cases has become an increasingly important research area in natural language processing (NLP) [1, 3, 10]. One approach to this task is Legal Judgment Prediction (LJP), where the goal is to predict a case outcome given the case facts [1]. In actual legal argumentation, professionals like lawyers and judges have extensive training and experience in legal reasoning, allowing them to make connections between case facts and relevant legal sources [10]. Casting LJP into an entailment task, where the goal is to classify if an article is violated or not violated given the case facts and the article (Figure 1), can enable the model to learn to reason between rules and case facts, improving its ability to predict legal outcomes [10]. In zero-shot learning, the goal is to train a machine learning model that learns to classify classes during testing that it has not seen during training. Zero-shot tasks are well-studied in deep learning for natural language processing (NLP) [8, 10, 12]. Zero-shot transfer is particularly interesting to the legal domain, where new articles and updates to legal codes and regulations are constantly being introduced [10]. The LJP entailment task can be used as a zero-shot transfer learning task by splitting articles into a source and target domain [10].

This paper focuses on the ECtHR dataset provided by LexGLUE [3] which can be transformed into an LJP zero-shot entailment task [10]. Authors of [10] have proposed a transformer-based entailment model to solve multiple ECtHR zero-shot LJP entailment task settings. A popular representation learning technique is to induce domain invariant representations using unsupervised target data and domain adversarial learning [5, 8, 10]. Authors of [10] have shown that domain adaptation techniques like the domain discriminator can improve model performance in both supervised and unsupervised domain adaptation in many different ECtHR zero-shot LJP entailment task settings. They have also conducted an experiment that showed performance increases when training on a related source domain. Mixture of Experts (MoE) is a technique used in machine learning where multiple models (referred to as "experts") are trained to solve a specific task, and their outputs are combined to make a final prediction. Previous work has shown that we can attempt to learn both domain-specific and shared representations from experts and general models respectively and combine their predictions [12]. We propose two MoE models with experts for each article in the source domain and a general model. The first model uses simple averaging of expert predictions. With our MoE attention model we attempt to utilize the benefits of training on related articles by aiming to learn to assign stronger weights to the predictions of experts of articles related to the input article. To the best of our knowledge, we are the first to apply a MoE approach to this task. Our goal is to show that our MoE models can increase the performance of the in [10] introduced entailment model with domain discriminator. Our MoE attention approach allows us to later analyze the weights given to each expert prediction deriving what the model learned about article-relatedness. We will also compare the attention model to the averaging model to see if a beneficial expert weighting strategy can be learned for unknown articles. We will further compare the weighting of the MoE attention model to what imagine to be related articles from a legal perspective.

Contributions: A novel MoE approach for zero-shot LJP entailment tasks

2 BACKGROUND AND RELATED WORK

Legal Judgement Prediction: Legal judgment prediction (LJP) is the task of using machine learning to predict the outcome of legal cases based on the presented facts. This area of research has gained significant attention in recent years due to its potential to assist judges, lawyers, and litigants in predicting the outcome of legal cases [3, 10]. LexGLUE provides a number of prominent LJP benchmark datasets including datasets with cases of the ECtHR and the US Supreme Court (SCOTUS) [3]. Despite recent progress, LJP still faces challenges such as limited data, bias in legal texts, and the need to account for different legal systems and languages [1, 3, 9].
Large pre-trained transformers, including legal-domain specific variants, have emerged as a popular approach for improving performance on natural language processing and LJP tasks [1, 2, 10].

Zero Shot Legal Judgement Prediction as Entailment Task: More recent work has proposed to transform the LJP task into an entailment task, where the goal is to classify whether a set of case facts violates a particular article given the facts and the article [10]. A benefit of the LJP entailment task is, that it can be used in zero-shot settings [10]. Authors of [10] transformed the ECtHR (Task A & B) datasets into an entailment dataset and conducted a number of experiments. They showed that the entailment task improves performance, especially for sparser articles, in comparison to the LJP classification task. In a number of zero-shot entailment settings, they were able to use domain adaptation methods (domain discriminators and Wasserstein distance) to increase performance, especially on the target domain. They have also found that training on related source articles 6 (right to a fair trial) and 8 (right to respect for private and family life) has better performance on target article P1-1 (the protection of property) compared to training on unrelated source articles 2 (right to life), 3 (prohibition of torture), and 5 (right to liberty and security). This relationship among articles was suggested by a legal expert. This experiment confirmed their intuition that the relatedness between source and target is to be considered when training a model for transferability. We want to utilize the performance increases of article-relatedness in our proposed MoE-Att model by training it to attend to experts of the articles in the source domain in a beneficial way depending on the input article. This should allow the model to increase performance on the target domain as the model is capable of weighting predictions of related experts higher compared to unrelated experts.

Mixture of Experts: Mixture of experts (MoE) is a technique that combines the predictions of multiple domain experts to produce a final prediction [6, 12]. While the expert combination does not need to occur at the prediction level, our proposed models use a similar approach. MoE models have been used for over 30 years and have shown benefits in regression and classification models, including support-vector machines, hidden Markov models, and neural networks like CNNs and RNNs [6, 13]. More recent work has demonstrated the benefit of using MoE in combination with large pre-trained transformer models and domain adaptation methods in zero-shot settings outside the legal domain [12]. We want to show that the entailment model introduced in [10], which is also based on large pre-trained transformer models, and their proposed domain adaptation methods can be outperformed when combined with an MoE approach.

3 DATASET, TASK & SETTING

We work with the ECtHR (Task A) dataset provided by LexGLUE [3], which consists of 11k case fact descriptions along with labels for articles the court has found to be violated. The dataset is split into 9k training (2001–2016), 1k validation (2016–2017), and 1k test (2017-2019) cases. The label set includes a subset of size 10 of prominent articles from the European Convention on Human Rights (ECHR). We follow the approach of [10] to transform this dataset into an entailment dataset in which given both the case fact statements and article information of one article, the model should classify whether this article has been found to be violated by the court. Equivalent to the authors of [10] we transform this into a zero-shot transfer task. The proposed approach can be viewed as an instance of domain adaptation, in which a "domain" corresponds to determining the outcomes of legal cases based on case facts with respect to a specific convention article [10]. In this study, 10 convention articles form 10 domains. The aim is to train a model on a source domain (the convention articles used for training) with the goal to perform well when tested on a target domain (unseen convention articles) [10]. To produce comparable results we follow the domain split proposed in [10]:

- *source domain*: 3, 5, 10, 14, 11
- *target domain*: 6, 8, P1-1, 2, 9

This is equivalent to their 1 to 0 transfer. We focus solely on a Zero-Shot Restrictive/Unsupervised Domain Adaptation (UDA) setting. In the UDA setting the model can be trained on labeled source data as well as unlabeled target data. Therefore, the objective is to train an entailment classifier using labeled data from the source domain, while also training on unlabeled target data trying to ensure that the model can perform well on the target domain. This setting is not uncommon in the legal domain, as new or modified legal rules are usually known for a given task and available for domain adaptation [10].

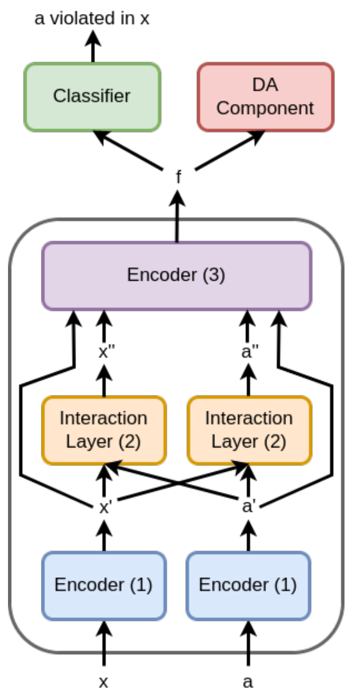

Figure 2: Entailment model architecture

4 METHOD

We employ a Mixture of Expert (MoE) model that consists of a general model and a domain-specific expert for each article in the source domain. The model takes case fact description x and article text a as input and outputs 1 if a is violated in x and 0 if not. The experts and the general model follow the entailment model architecture proposed in [10]. For the general model, we use a domain discriminator as domain adaptation component. This makes each expert's architecture equivalent to that of the source-only model introduced in [10] and the general model architecture equivalent to that of their entailment model with domain discriminator (DD) architecture.

4.1 Entailment Model

As mentioned our architecture is equivalent to the entailment model architecture (Figure 2) proposed in [10]. The entailment model consists of a feature extractor F, a classifier C, and a domain adaptation (DA) component D (Figure 2). In our MoE model, only the general model has a domain discriminator as its' DA component. The feature extractor consists of three layers. Encoding layer (1) uses LegalBERT [2] with frozen weights for sentence-wise encoding, followed by an attention layer creating sentence-level representations passed to a GRU encoder generating context-aware sentence representations x' and a'. Interaction layer (2) creates a fact-aware article representation a'' and an article-aware facts representation x'' from the sentence representations x' and a' via dot product attention between the two sequences of sentences. Encoding layer (3) combines x', x'', a', a'' to produce a final article-dependent representation f of case facts x. Encoding f is then fed to the classification layer which classifies if a is violated in x.

The goal of DA is to train a model that develops the ability to generalize - reason about article violation in a case similar to a lawyer - when presented with unknown articles [10]. To achieve such generalization one can minimize the textual information encoded in f with the DA component [10]. With the DA component, we can enable feature vector f encoding *violation* or *no violation* with as little information about the article as possible. The goal of the DA component is ensuring that f produced given x, a does not encode a.

4.2 Domain Discriminator

The goal of using a domain discriminator (DD) during training with is to minimize the ability to predict the domain from the feature representation of the input [5]. Using a gradient reversal layer (GRL) between the feature extractor and discriminator allows using stochastic gradient descent (SGD) based optimization strategies which minimize the prediction loss L_y and the domain prediction loss L_D [5, 10]. During the backward pass, the feature extractor receives the opposite gradients of the domain discriminator scaled by hyperparameter λ from the GRL resulting in opposite gradient updates [5, 10]. During the forward pass the GRL acts as the identity function [5]. Adapted to the LJP entailment task the objective function in [10] is:

$$L_D = CELoss(D(GRL(F(x,a))), y_d)$$

$$L_y = BCELoss(C(F(x,a)), y_e)$$

$$\arg \min_{\theta_F, \theta_C, \theta_D} L_y + \lambda L_D$$

with y_d the expected discriminator output, a one-hot vector of length ten for classifying the input article, and $y_e \in \{0, 1\}$ for the entailment label. F, C, and D represent feature extractor, classifier, and discriminator with parameters θ_F, θ_C, θ_D respectively. While we can train the entire entailment architecture on labeled source data $\{(x, a, y_e) | a \in A, x \in F, y_e = a\, violated\, in\, x\}$, we can additionally train the model on unlabeled target data $\{(x, a) | a \in A', x \in F\}$ by excluding the classifier.

4.3 Batch sampling strategy

When transforming the $9k$ training cases of the ECtHR (Task A) dataset into an entailment task we get $9k \cdot 10 = 90k$ entries from which $45k$ belong to the source domain. From these $45k$ entries around $41.7k$ are labeled with 0 and only around $3.3k$ are labeled with 1. This dataset imbalance poses a challenge as a balanced training set is required (an equal number of cases with and without violations) to avoid bias towards a specific class [1]. To achieve balance during training we apply random oversampling (duplicating random entries in the minority class) as well as random undersampling (not using random entries of the majority class) in each epoch. Research shows that oversampling generally leads to better results than undersampling for different machine learning classifier models [13]. Each epoch contains $\lfloor 45k/batchsize \rfloor$ entries and each batch has a balanced number of 0 and 1 entries for each article. In MoE each batch has only entries for the same article to allow joint training of the experts and the general model. To assure entries of each article are trained with the same frequency, a random article from the source domain is chosen for each batch. Additionally, we

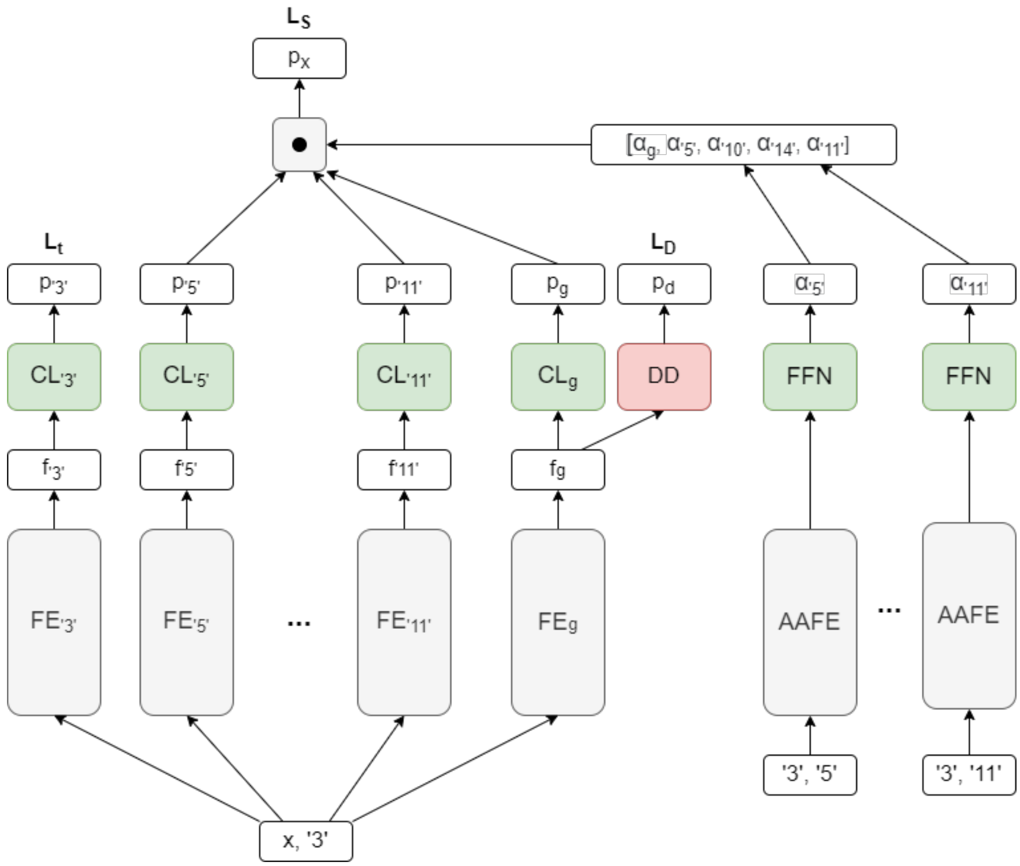

Figure 3: MoE-Att model architecture: The image showcases a training scenario with input pair (x, '3'). Article '3' is the target article; we compute loss L_t for E_3. L_S is the loss for the combined predictions of the experts of the remaining articles and the general model. L_D is the domain discriminator loss.

have significant discrepancies in the occurrence of violations between domains, such as 1349 violations for article '3' in the training dataset compared to only 110 for article '11'. As a result, the positive entries for article '11' will be duplicated much more frequently compared to those of article '3'.

4.4 Prediction Level MoE Models

We employ a Mixture of Expert (MoE) model that consists of a general model (GM) and a domain-specific expert for each article in the source domain S (Figure 3). The GM consists of a feature extractor FE_g, classifier CL_g, and domain discriminator DD; the general model architecture is equivalent to the entailment model with DD in [10]. Each expert E_i with $i \in$ *source domain* consists of a feature extractor FE_i and a classifier CL_i. The expert architecture is equivalent to the architecture of the entailment model without DA component in [10]. The expert and general model predictions are weighted according to the vector $\alpha = [\alpha_g, \alpha_3, \alpha_5, \alpha_{10}, \alpha_{11}, \alpha_{14}]$ and combined to a final prediction p_x. We employ two different models, MoE-Avg which keeps α constant, and MoE-Att which produces each α_i for expert E_i with an additional model from the input article text a and the text for expert article i. The general function to produce the final prediction p_x is:

$$p_x(x, a, S) = \alpha_g \cdot GM(x, a) + (1 - \alpha_g) \cdot \sum_{i \in S} \alpha_i \cdot E_i(x, a)$$

We follow a similar source training approach to the training strategy proposed in [12]: For each batch, the batch sampler provides balanced entries for a target article t from the source domain, with the rest of the articles from the source domain treated as meta-sources $S' = S \setminus \{t\}$. Two losses are calculated, namely L_t for the expert of the target article and L_S a loss for the remaining experts and the GM:

$$L_t = BCELoss(E_t(x, a))$$

$$L_S = BCELoss(p_x(x, a, S'))$$

This results in the combined loss:

$$L = L_t + L_S + L_D$$

where L_D is the discriminator loss of the general model. When training on unlabeled target data, we can only train the general model without its classifier, as the experts cannot be trained on unlabeled data.

At test time, if the input article is from the source domain, the prediction of the according expert is taken as the final prediction, only if the input article is from the target domain the final prediction is

Table 1: Task A, F1 performance of source only, domain discriminator, and MoE models

Setting	Model	mac.	mic.	mac.	mic.
Baseline	Source only	67.79	74.57	5.80	8.02
UDA	DD	**68.19**	**75.32**	**14.47**	**16.51**
	MoE-Avg	66.10	74.04	5.03	8.68
	MoE-Att	66.84	74.29	7.05	12.04

calculated from $p_x(x, a, S)$.

MoE-Avg For this model we chose a constant alpha with $\alpha_g = 0.5$ and $\alpha_i = \frac{1}{|S|}$.

MoE-Att This model is far more interesting to us as its architecture allows the model to learn to attend to its' experts differently depending on the input article. As mentioned before, authors in [10] found performance increases when training on a related source domain compared to an unrelated source domain. The idea behind MoE-Att is to utilize this benefit by learning to weigh experts in a beneficial manner to increase overall performance. To allow the model to do so we train an additional model within our MoE-Att architecture (Figure 3) that produces each α_i for expert E_i from the input article a and the expert article i. The model producing the α_i consists of an article-article feature extractor (AAFE) and a feed-forward network (FFN). The AAFE is a modified version of the FE of the entailment model. The encoding layer (1) and interaction layer (2) stays identical. For input pair (a_1, a_2) the encoding layer (3) however, does not produce an a_2-dependent version of a_1 like the encoding layer (3) in the entailment FE would, but a concatenation of a (a_1', a_2') where a_1' is a a_2-dependent version of a_1 and a_2' a a_1-dependent version of a_2. The FFN then produces the α_i. We apply a $softmax$ to all α_i with $i \in S$. This model learns how to attend to experts depending on the input article and expert articles. This model, therefore, learns which "article-relatedness" is beneficial during training time for articles of the source domain and can hopefully transfer that to articles of the target domain at test time. We will investigate the article relatedness this model learns in section 5.2.

5 EXPERIMENTS & DISCUSSION

All model implementations can be found on GitHub[1]. Detailed hyperparameters for all models are described in Appendix Sec. A. The source-only model and DD model results are from paper [10] to make comparability to the paper easier - we were able to produce very similar results for Task A, UDA, $1 \longrightarrow 0$ setting for their source only and DD model. Our MoE models are also trained and tested in this setting.

5.1 Can MoE increase performance compared to the DD model?

Table 1 summarizes the results of our two experiments comparing the MoE models to the DD and source-only models of paper [10]. Our findings show that the trained MoE models did not surpass the DD model in performance. The MoE-Avg model produced results comparable to the baseline source-only model, while the MoE-Att model outperformed the baseline but still fell short of the DD model. However, we believe that our MoE models' low performance on the target domain might be attributed to the constant lambda hyperparameter used in the GRL in our experiments. This approach differs from the changing lambda strategy proposed in [4], which was used in [10]. As we only had a single attempt to train our MoE models due to the increase in size and the resulting increase in training time compared to the DD model further experimentation and adjustments to the hyperparameters may be necessary to achieve better results. While our experiments did not show improvements with MoE over the DD model, further exploration and optimization may reveal the potential benefits of incorporating MoE in the LJP zero-shot entailment task.

5.2 Can MoE-Att utilize article relatedness?

Firstly, it is worth noting that the MoE-Att model outperformed MoE-Avg, particularly in the target domain. This indicates that the α_i weights produced by our model were able to weight the experts in the target domain in a beneficial manner, suggesting that our model was able to learn beneficial article relatedness even for unknown articles in the target domain.
To gain insights into how the MoE-Att model weights each expert depending on the input article, we plot the α values in Figure 4. It is worth noting that the article-relatedness that is useful to our MoE model may differ from the articles that legal experts view as related. Interestingly, for most input articles, the model chooses a single expert, which may or may not be desirable. We could introduce a penalty to discourage this behavior in future work if this behavior was not desired.
Since we are not legal experts, we use articles that frequently occur together in our dataset (Figure 5) as an estimate for possible relatedness. We find that articles 3 and 5 occur together very frequently in the training dataset, and the model chooses expert 3 for input article 5 and expert 5 for input article 3 during training. However, for input articles 10, 14, and 11 of the source domain, we do not observe such a correlation. These articles occur much less frequently in the dataset, and Figure 5 may not be a reliable indicator of their relatedness from a legal perspective. We can also observe that expert 10 is the most frequently chosen expert. We also observe that expert 10 is the most frequently chosen expert, despite article 10 occurring infrequently in the dataset. This suggests that expert 10 may have learned to capture certain aspects of legal reasoning that are useful for the other articles in the source domain. Overall, it is challenging to determine if there is a correlation between the produced α values (i.e., article-relatedness useful to the model) and the articles that legal experts consider related, without consulting a legal expert.

[1]https://github.com/KorbiQWeidinger/ljp-entailment

	exp-3	exp-5	exp-10	exp-14	exp-11
3	nan	1.0	0.0	0.0	0.0
5	1.0	nan	0.0	0.0	0.0
10	0.0	0.0	nan	0.0	1.0
14	0.0	0.0	1.0	nan	0.0
11	0.0	0.0	1.0	0.0	nan
6	1.0	0.0	0.0	0.0	0.0
8	0.0	0.0	1.0	0.0	0.0
P1-1	0.0	0.0	1.0	0.0	0.0
2	0.1	0.9	0.0	0.0	0.0
9	0.0	0.0	1.0	0.0	0.0

Figure 4: $softmax([\alpha_3, \alpha_5, \alpha_{10}, \alpha_{14}, \alpha_{11}])$ produced by MoE-Att

	3	5	10	14	11
3	nan	454	7	26	12
5	454	nan	6	2	15
10	7	6	nan	0	3
14	26	2	0	nan	5
11	12	15	3	5	nan
6	196	259	37	15	16
8	105	107	5	61	1
P1-1	22	16	1	26	0
2	222	199	0	5	1
9	4	2	0	10	2

Figure 5: The frequencies at which articles occur together in the ECtHR Task A training dataset

6 CONCLUSION

Our findings suggest that MoE may not always be the best choice for improving zero-shot transfer learning performance. While MoE has shown promising results in other NLP tasks, such as machine translation and language modeling, our experiments suggest that it may not be effective for improving zero-shot transfer learning performance in the specific context of legal entailment. It is possible that the constant lambda hyperparameter used in the GRL during our experiments might have limited the effectiveness of the MoE models in learning useful domain-invariant features. Further experimentation with varying lambda values and other hyperparameters may be necessary to confirm this hypothesis and achieve better results. Our study provides some preliminary evidence that MoE-Att can utilize article-relatedness to improve performance on unknown articles in the target domain. However, further investigation and consultation with legal experts are necessary to confirm whether the article-relatedness learned by MoE-Att corresponds to what legal experts would consider being related articles.

7 LIMITATIONS AND FUTURE WORK

Besides the limitations of the underlying entailment model listed in [10], the main limitation of MoE compared to for example the DD of [10] is the large increase in model size. Due to limited resources and the large amount of time it takes to train these big models, we were only able to train each model for a single time. As there still were some misconfigurations and training such models often takes multiple tries due to hyperparameter tuning we believe that the numbers presented in table 1 are not representative of the actual capability of our proposed MoE models.

As the misconfigurations have been identified and fixed, future work can explore the changing lambda strategy proposed in [4], which might help improve the performance of our MoE models.

The proposed models can additionally be trained and tested in many more scenarios on the ECtHR entailment dataset as well as other datasets. Future work could also investigate a MoE architecture that combines feature representations instead of predictions. To reduce the overall model size reducing the size of the experts is a potential strategy. Furthermore, our experiments were conducted on a single GPU and did not utilize distributed training. Future work can investigate the performance gains that can be achieved by distributing the training process across multiple GPUs using for example domain expert mixture layers [7]. Recent work has also shown promising results with hypernetworks in zero-shot tasks [11] which are smaller in size compared to MoE.

Lastly, while our MoE-Att model demonstrated the ability to weigh experts based on input articles, it remains unclear if this article-relatedness aligns with legal expert opinions. Future work can consult legal experts to determine if the produced α values are correlated with actual article-relatedness in the legal domain.

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A IMPLEMENTATION DETAILS

Our MoE models script can be found on GitHub[2]. Both MoE models were executed with a constant learning rate of 0.0001, a batch size of 16, and an α_g of 0.5. The lambda in the GRL was kept constant at 0.2. The size of each feature extractor layer as well as the AAFE is equivalent to the size of each feature extractor layer in [10]. The dropout rate is 0.1 in all layers like in [10].

[2]https://github.com/KorbiQWeidinger/ljp-entailment/blob/master/src/moe_pred.py